

HOUSEHUNT – PROJECT REPORT

Team ID: LTVIP2026TMIDS65007

Project Type: MERN Full Stack Web Application

1. INTRODUCTION

1.1 Project Overview

HouseHunt is a full-stack web application built with the **MERN (MongoDB, Express.js, React.js, Node.js)** stack. It serves as a comprehensive property rental platform designed to connect renters and property owners within a secure, centralized ecosystem.

1.2 Purpose

To simplify the house-hunting process by providing an intuitive interface for **Renters** to find homes, for **Owners** to manage their property listings, and for **Admins** to oversee platform security and user approvals.

2. IDEATION PHASE

2.1 Problem Statement

Current rental processes are often fragmented, lacking a dedicated platform that offers a seamless transition from property browsing to lease negotiation and booking management with integrated administrative oversight.

2.2 Empathy Map Canvas

- **Think & Feel:** Frustrated by the difficulty of finding verified rental properties and reliable owners.
- **See:** Disorganized property listings and lack of clear status updates on rental inquiries.
- **Say & Do:** "It's hard to trust online listings or know if a property is still available".
- **Hear:** "It's hard to trust online listings or know if a property is still available".

- **Pain:** Complex booking processes and a lack of verified owner profiles.
- **Gain:** A transparent, secure environment for property transactions and real-time status tracking.

2.3 Brainstorming Ideas

- **Role-based dashboards** for Renters, Owners, and Admins.
 - **JWT-based authentication** for secure login and registration.
 - **CRUD functionality** for owners to manage property listings (Add/Edit/Delete).
 - **Advanced search filters** for location, rent range, and bedroom count.
 - **Admin approval system** for owner registration and platform governance.
 - **In-app messaging** for lease negotiation and agreement finalization.
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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Renter: Register → Filter/Browse Properties → Send Inquiry → Negotiate Lease → Confirm Booking.

Owner: Register (Pending Admin Approval) → List Properties → Manage Availability → Respond to Inquiries.

Admin: Monitor Platform → Approve Owners → Maintain Trust & Safety.

3.2 Solution Requirements

- **Functional roles:** Renter, Owner, and Admin.
- **Secure Access:** Protected routes and APIs using **JWT**.
- **Database:** **MongoDB** for persistent storage of user profiles and property data.
- **Interface:** Responsive UI using **Bootstrap** and **Material UI**.

3.3 Data Flow Diagram

[Frontend - React] → API Call → [Backend - Express Server]

→ Token & Role Validation → [MongoDB Database]

3.4 Technology Stack

- **Frontend:** React.js, Axios
 - **Backend:** Node.js, Express.js
 - **Database:** MongoDB (Compass)
 - **Authentication:** JWT
 - **UI & Charts:** Custom CSS + Recharts
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4. PROJECT DESIGN

4.1 Problem-Solution Fit

HouseHunt addresses the lack of transparency in rentals by enabling direct interaction between renters and verified owners, supported by admin moderation to ensure a safe marketplace.

4.2 Proposed Solution

- **Centralized Rental Marketplace:** One platform for all property types.
- **Owner Management Suite:** Tools for landlords to track occupancy and edit listings.
- **Admin Governance:** Oversight of user registrations to prevent fraudulent listings.
- **Booking Pipeline:** Clear status tracking from "Inquiry" to "Move-in".

4.3 Solution Architecture

- **Component-based React frontend** for a responsive UI.
 - **Express.js server** to handle business logic and communication.
 - **Role-based rendering** to show different dashboards to Renters and Owners.
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5. PROJECT PLANNING & SCHEDULING

5.1 Project Timeline

Phase	Duration
Requirements Gathering	2 Days
UI/UX Design	3 Days
Backend Development	4 Days
Frontend Integration	4 Days
Testing & Debugging	2 Days
Deployment	1 Day

6. FUNCTIONAL & PERFORMANCE TESTING

6.1 Performance Testing

- **API Response Time:** Latency maintained at < **300ms** using **Axios** and **RESTful APIs** for rapid client-server exchange.
 - **Database Queries:** Optimized **MongoDB** search filters for location, rent, and bedrooms to ensure high-speed data retrieval.
 - **Token Middleware:** **JWT-based authorization** secures routes across roles (Renter, Owner, Admin) with minimal overhead.
 - **Manual Testing:** Validated registration, property CRUD, and inquiry workflows via **Thunder Client**.
 - **Database Inspection:** Verified integrity of user profiles and booking data using **MongoDB Compass**.
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7. RESULTS

7.1 Output Screenshots

HouseHunt

LoginRegister

Find Your Home

Location:

e.g., Mumbai, Delhi

Min Price (₹/month):

0

Max Price (₹/month):

Any

Bedrooms (BHK):

Any

Property Type:

Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

Location: tirupati

₹15,000/month

3 BHK (2 Bath(s)) 10000 sq ft

Type: Apartment (Furnished: Unfurnished)

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK (2 Bath(s)) 1500 sq ft

Type: Apartment (Furnished: Unfurnished)

Posted by: Pavan (Listed as: Owner)



Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK (1 Bath(s)) 183539 sq ft

Type: House (Furnished: Unfurnished)

Posted by: vamsi (Listed as: Owner)

HouseHunt

LoginRegister

Find Your Home

Location:

e.g., Mumbai, Delhi

Min Price (₹/month):

0

Max Price (₹/month):

Any

Bedrooms (BHK):

Any

Property Type:

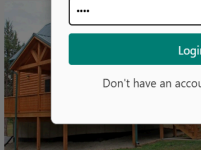
Any

Furnished Status:

Any

Listed By:

Available Rentals



Royal house

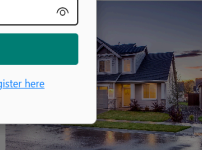
Location: tirupati

₹15,000/month

3 BHK (2 Bath(s)) 10000 sq ft

Type: Apartment (Furnished: Unfurnished)

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK (2 Bath(s)) 1500 sq ft

Type: Apartment (Furnished: Unfurnished)

Posted by: Pavan (Listed as: Owner)



Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK (1 Bath(s)) 183539 sq ft

Type: House (Furnished: Unfurnished)

Posted by: vamsi (Listed as: Owner)

Login

Username:

Vamsi

Password:

....

Login

Don't have an account? [Register here](#)

Furnished Status:

Any

Listed By:

Any

Search Homes

Reset Filters

Post New Ad

Type: Apartment (Furnished: Unfurnished)

Posted by: Royal (Listed as: Owner)

this is Royal house



Spacious 3 BHK in Koramangala

Location: Bangalore

₹65,000/month

3 BHK (3 Bath(s)) 1800 sq ft

Type: Apartment (Furnished: Semi-Furnished)

Posted by: rajeshK (Listed as: Owner)

Type: Apartment (Furnished: Unfurnished)

Posted by: Pavan (Listed as: Owner)

this is pavan house



Cozy 1 BHK Studio, Andheri West

Location: Mumbai

₹32,000/month

1 BHK (1 Bath(s)) 650 sq ft

Type: Apartment (Furnished: Furnished)

Posted by: priyaS (Listed as: Owner)

Type: House (Furnished: Unfurnished)

Posted by: vamsi (Listed as: Owner)

this is my house. please call me



Independent House, Jubilee Hills

Location: Hyderabad

₹90,000/month

4 BHK (4 Bath(s)) 3200 sq ft

Type: House (Furnished: Unfurnished)

Posted by: amitV (Listed as: Owner)

Well-ventilated 3 BHK apartment with modern amenities, car parking, and close to tech parks. Ideal f...



Modern 2 BHK, Powai Lake View

Location: Mumbai

₹72,000/month

2 BHK (2 Bath(s)) 1150 sq ft

Type: Apartment | **Furnished:** Furnished

Posted by: rajeshK (Listed as: Dealer)

Luxurious 2 BHK in a premium complex with lake view. Includes modular kitchen and society amenities.

Fully furnished studio apartment perfect for singles. Walking distance to Andheri station. Includes ...



Builder Floor 3BHK, GK-1

Location: Delhi

₹58,000/month

3 BHK (2 Bath(s)) 1600 sq ft

Type: Condo | **Furnished:** Semi-Furnished

Posted by: sunitaM (Listed as: Builder)

Newly constructed builder floor with lift and stilt parking. Spacious rooms and good quality fitting...

Large independent house with a beautiful private garden and ample parking. Prime, peaceful locality.

HouseHunt

Logged in as: Vamsi [Logout](#)

Find Your Home

Location:

Min Price (₹/month):

Max Price (₹/month):

Bedrooms (BHK):

Property Type:

Furnished Status:

Listed By:

Available Rentals



Royal house

Location: tirupati

₹15,000/month

3 BHK (2 Bath(s)) 10000 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Royal (Listed as: Owner)



Pavan house

Location: Tirupati

₹15,000/month

2 BHK (2 Bath(s)) 1500 sq ft

Type: Apartment | **Furnished:** Unfurnished

Posted by: Pavan (Listed as: Owner)

HouseHunt

[Login](#) [Register](#)

Find Your Home

Location:

Min Price (₹/month):

Max Price (₹/month):

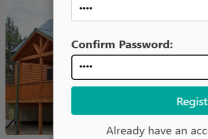
Bedrooms (BHK):

Property Type:

Furnished Status:

Listed By:

Available Rentals



Royal house

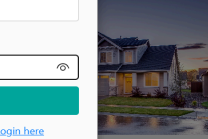
Location: tirupati

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Vamsi House

Location: Kodur

₹5,00,000/month

4 BHK (1 Bath(s)) 183539 sq ft

Type: House | **Furnished:** Unfurnished

Posted by: vamsi (Listed as: Owner)

✕

Register

Username:

Password:

Confirm Password:

[Register](#)

Already have an account? [Login here](#)

8. ADVANTAGES & DISADVANTAGES

✓ Advantages

- **Direct Communication:** Integrated messaging for lease negotiations.
- **Secure Framework:** Role-based access and JWT protection.
- **Scalable Architecture:** Easily expandable to include more property categories.
- **User Transparency:** Real-time dashboard updates for booking statuses.

✗ Disadvantages

- **Manual Validations:** Current reliance on admin for manual owner approval.
- **Local Storage:** Currently limited to URL-based property images rather than cloud storage.
- **Limited Reach:** Currently deployed in a local development environment.

9. CONCLUSION

HouseHunt successfully streamlines the rental journey by providing a MERN-based solution that balances the needs of renters and owners. The platform demonstrates how centralized governance and role-based access can create a more trustworthy and efficient property market.

10. FUTURE SCOPE

- **Cloudinary Integration:** For direct property image uploads.
 - **Payment Gateway:** Integration of Stripe or Razorpay for secure rent deposits.
 - **Virtual Tours:** 360-degree viewing of properties within the app.
 - **Map Integration:** Using Google Maps API for precise property location tracking.
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11. APPENDIX

- **Source Code:** <https://github.com/manikya126/HouseHunt-Project>

- **Demo Video:**

https://drive.google.com/file/d/1usM-gywIgUdo-oDLOxneCJxn2rnmC_Q4/view?usp=drivesdk